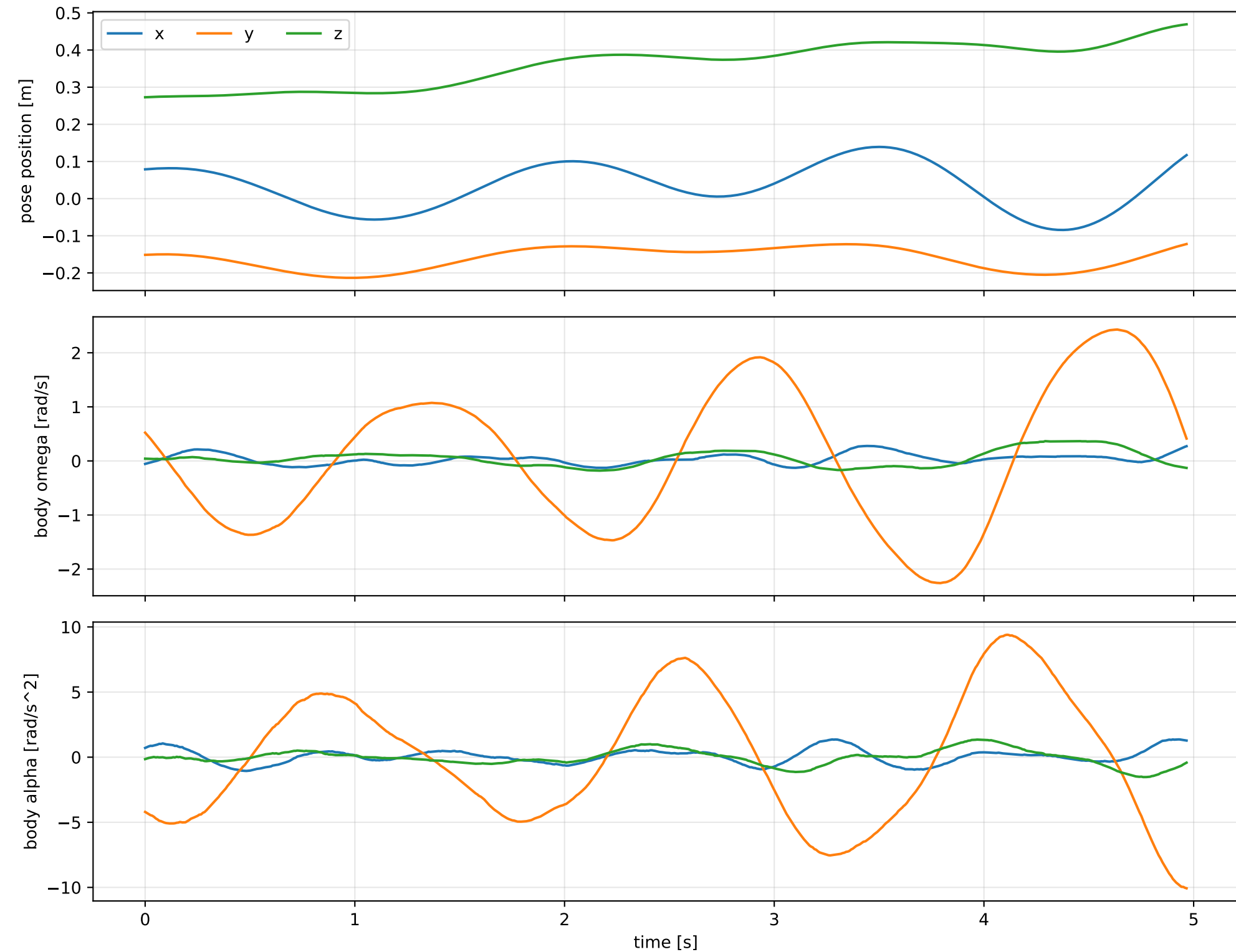
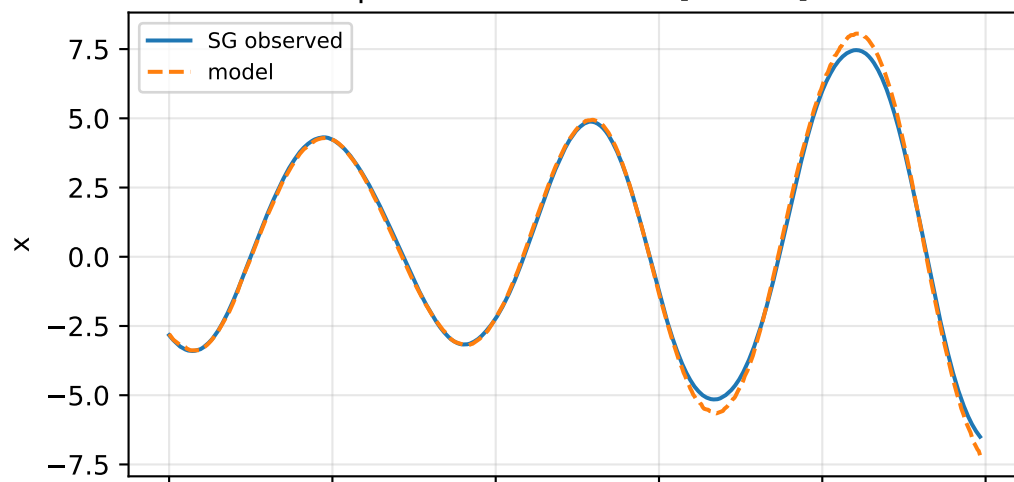


cov_diagonal: SG trajectory and derived motion

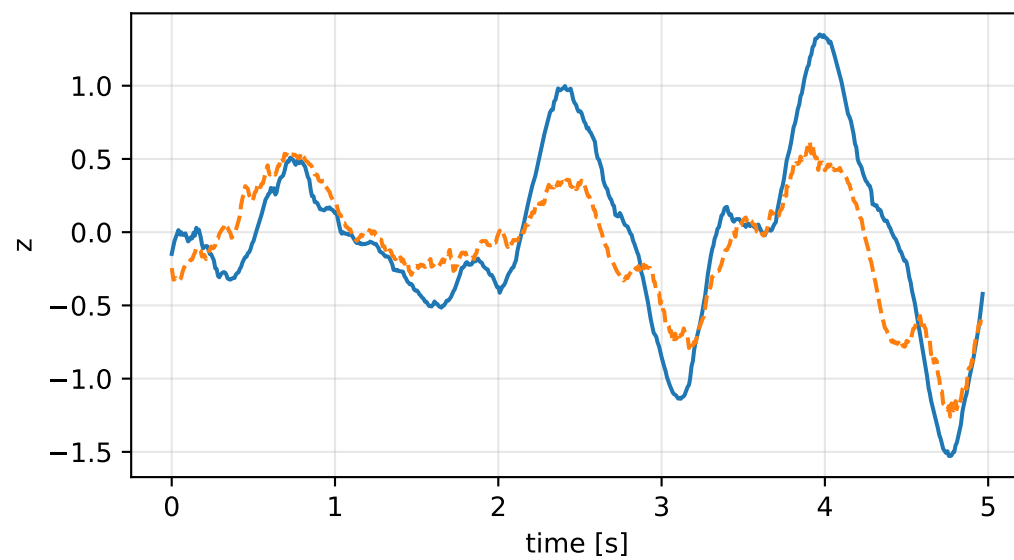
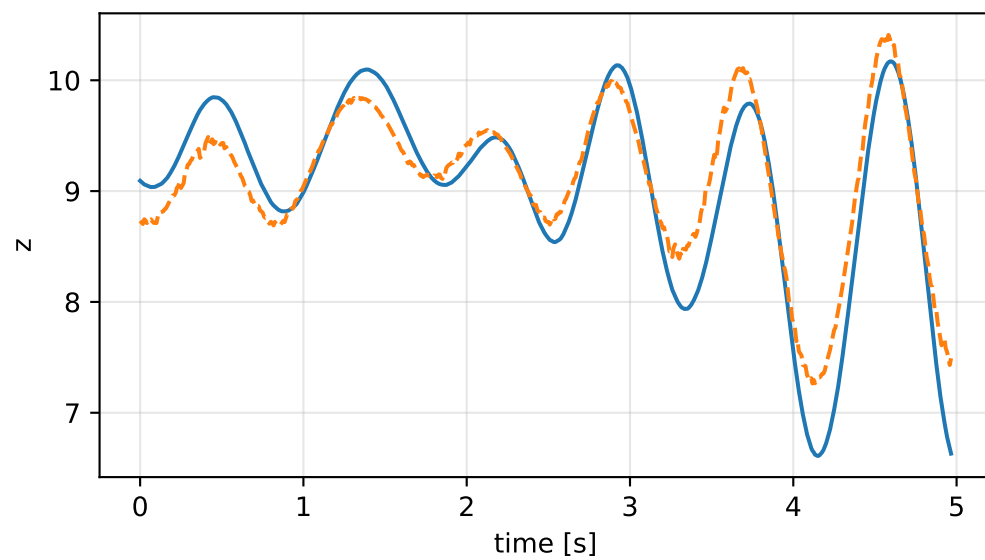
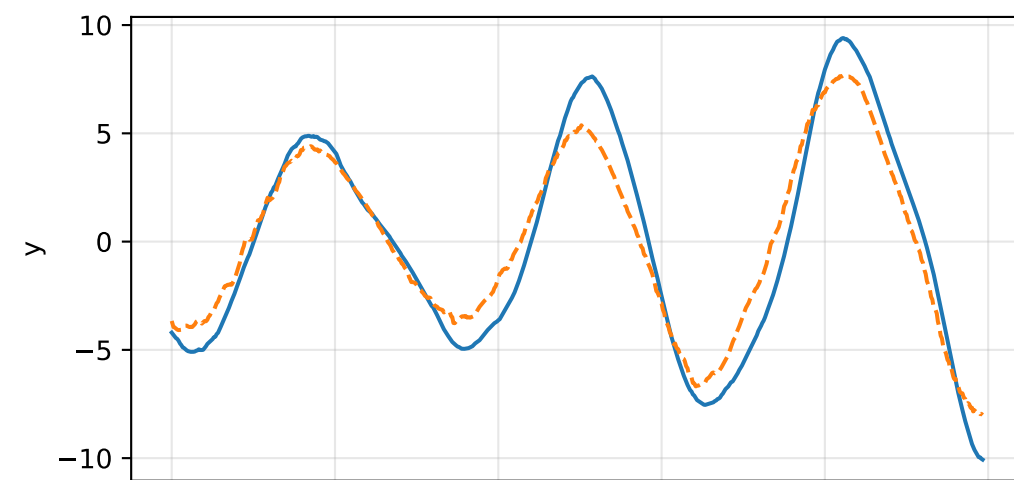
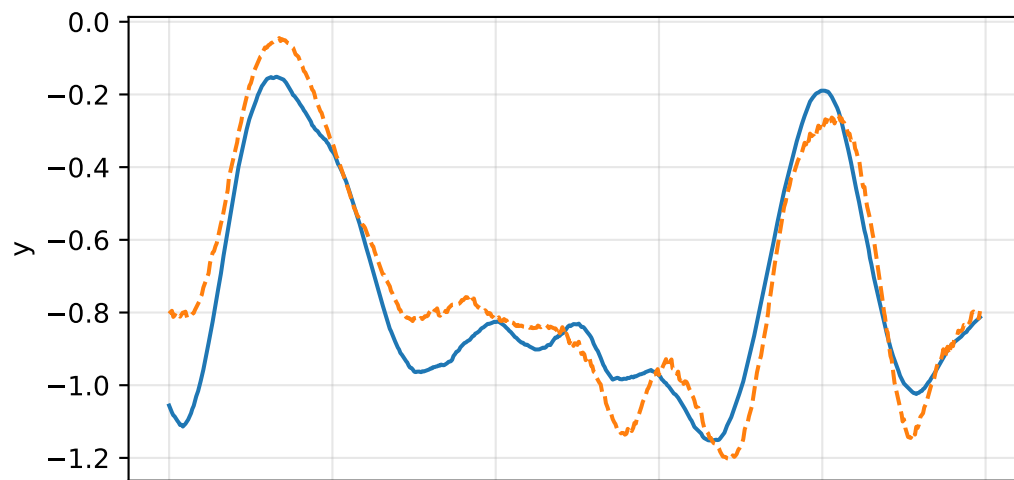
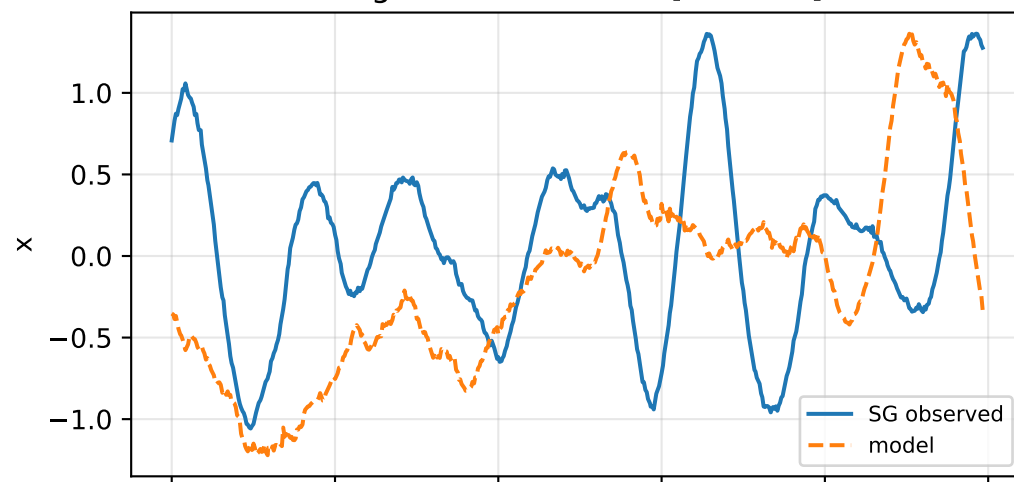


cov_diagonal: acceleration residual objective

specific acceleration [m/s²]

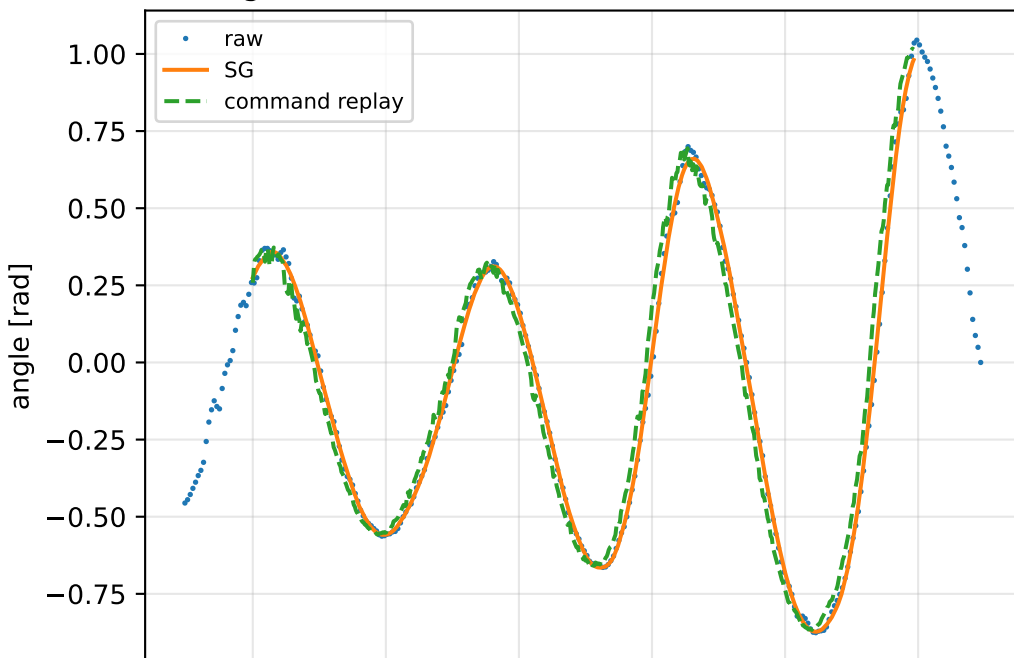


angular acceleration [rad/s²]

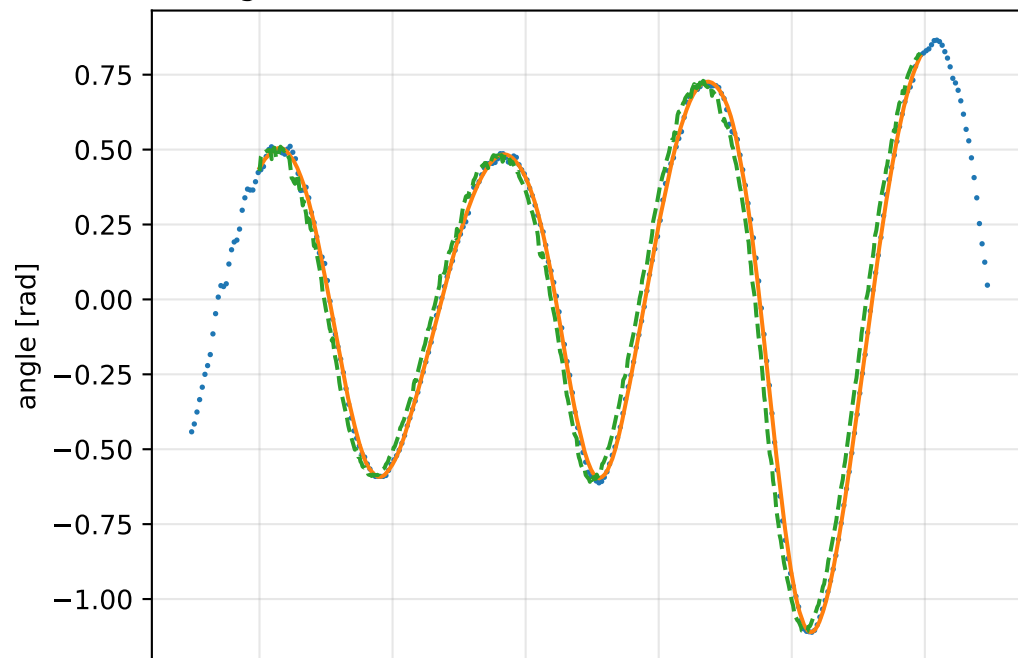


cov_diagonal: gimbal measurement audit (objective=measured_sg)

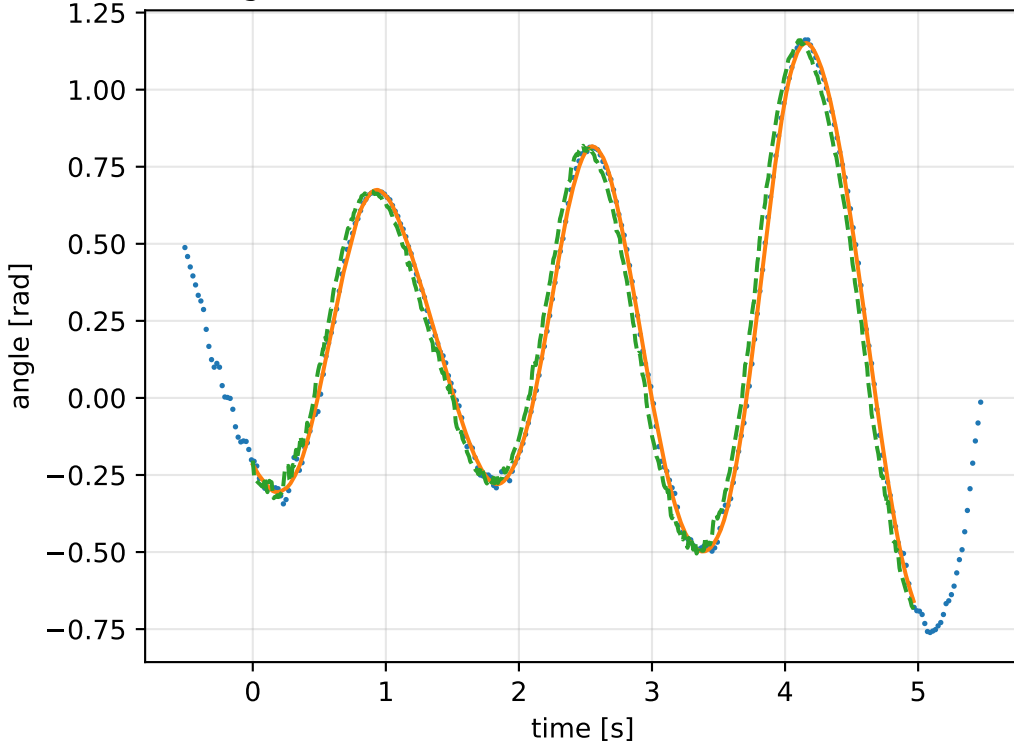
gimbal 1: RMSE=0.0759, max=0.2095 rad



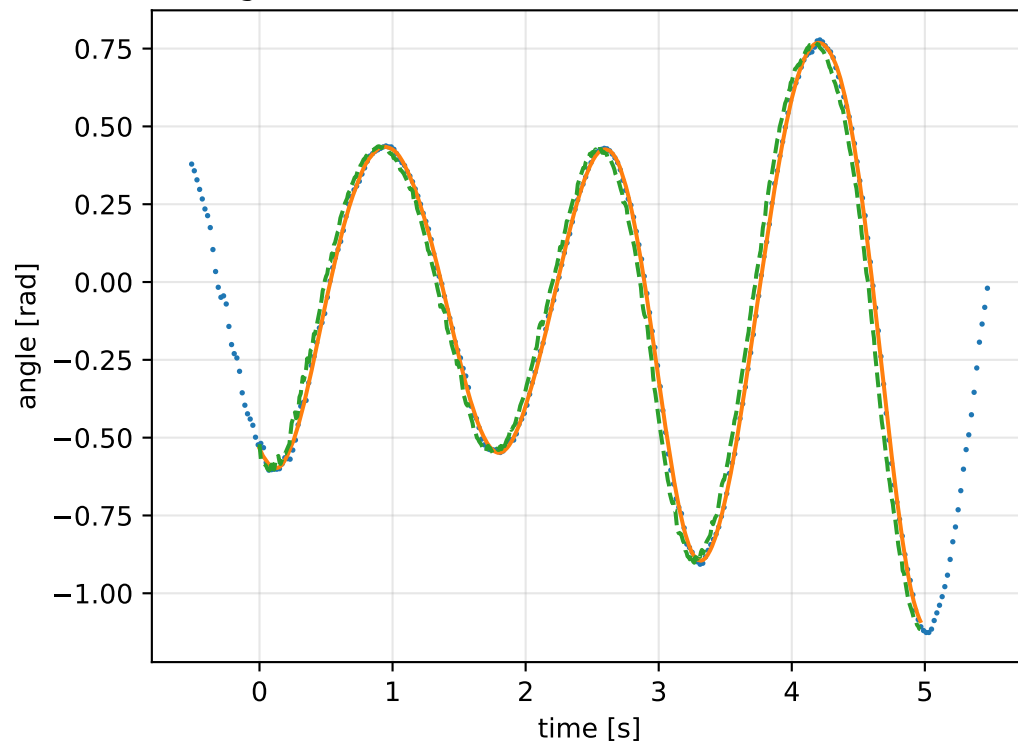
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

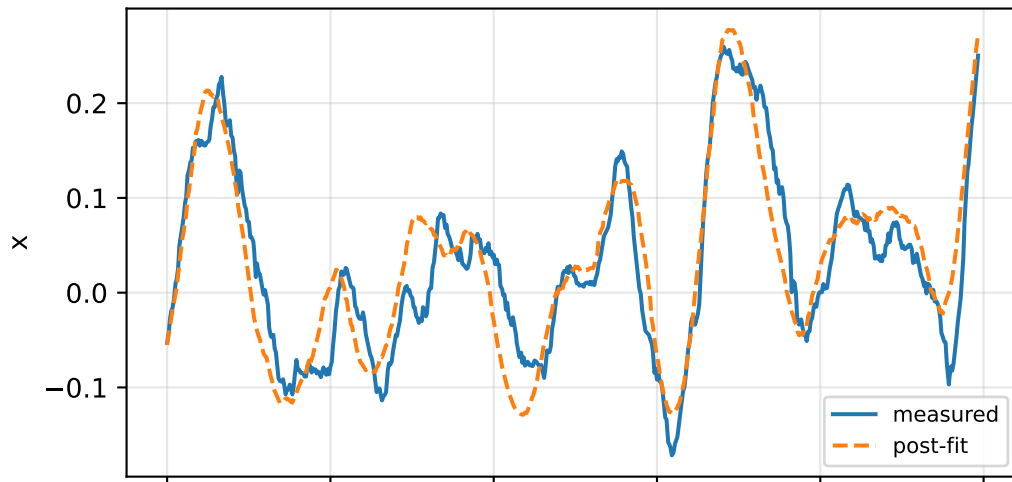


gimbal 4: RMSE=0.07131, max=0.1737 rad

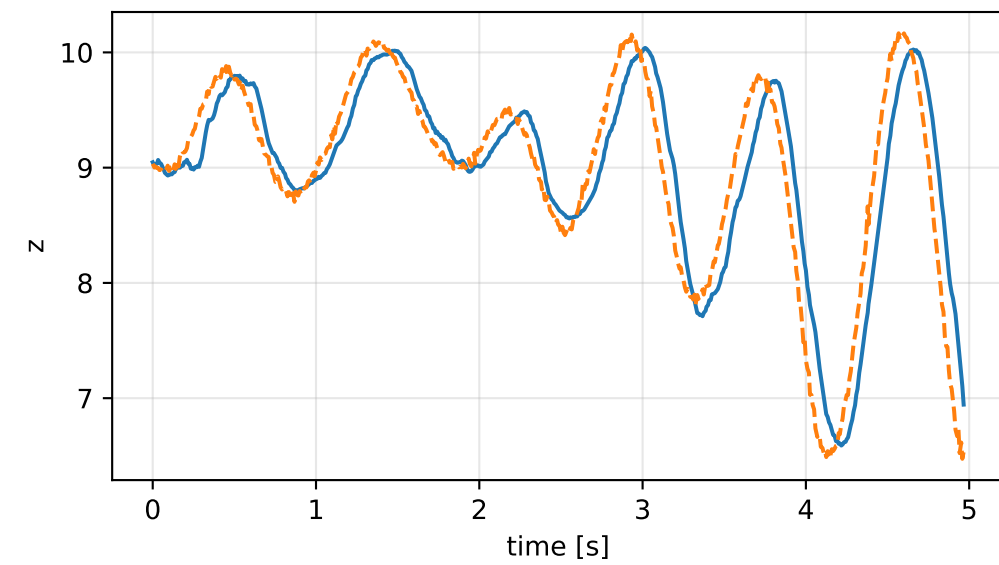
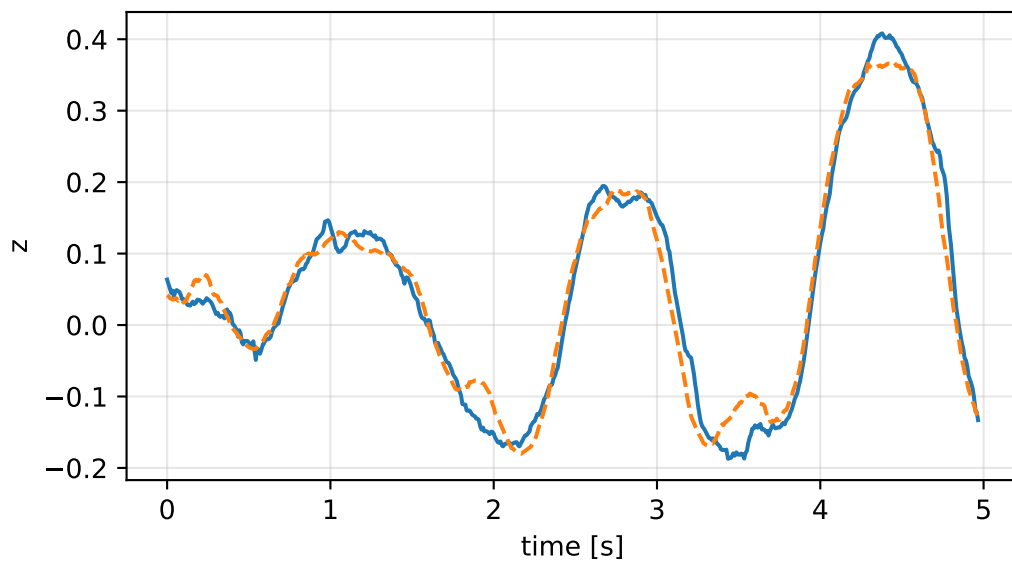
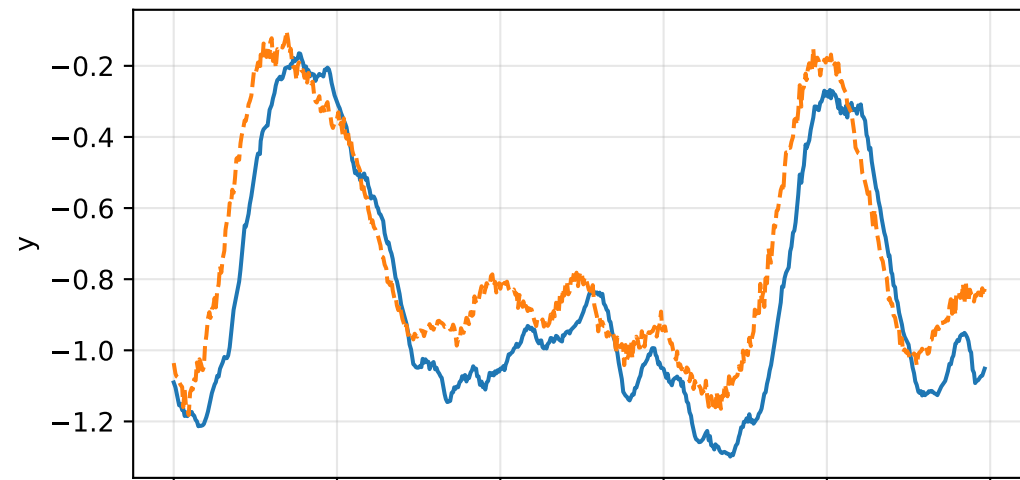
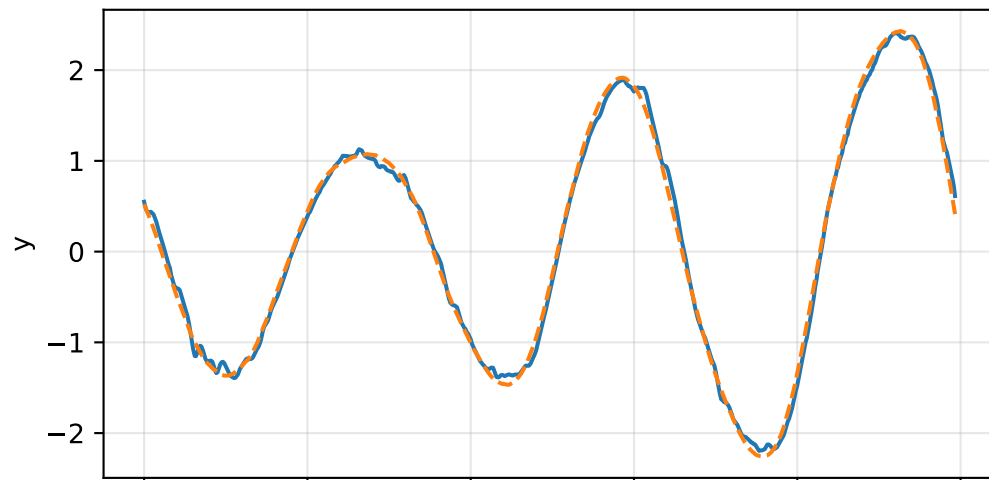
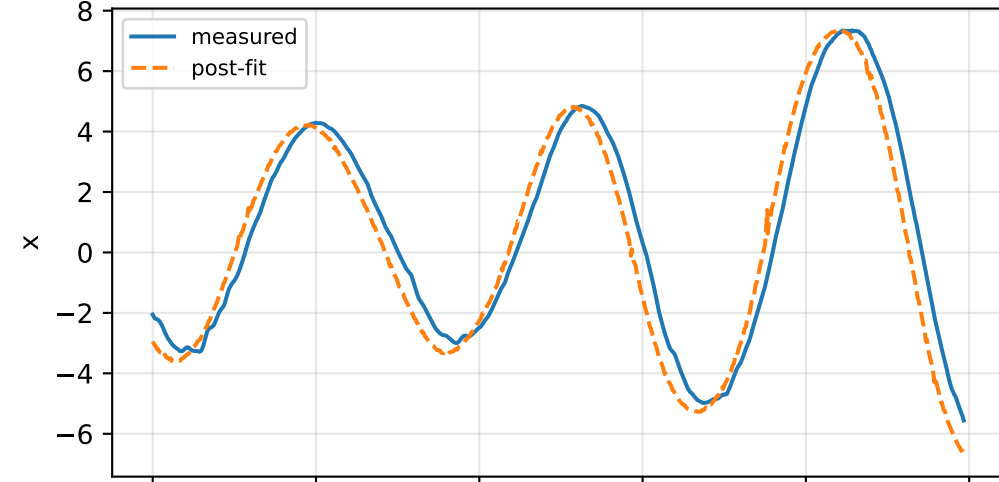


cov_diagonal: IMU comparison (diagnostic only)

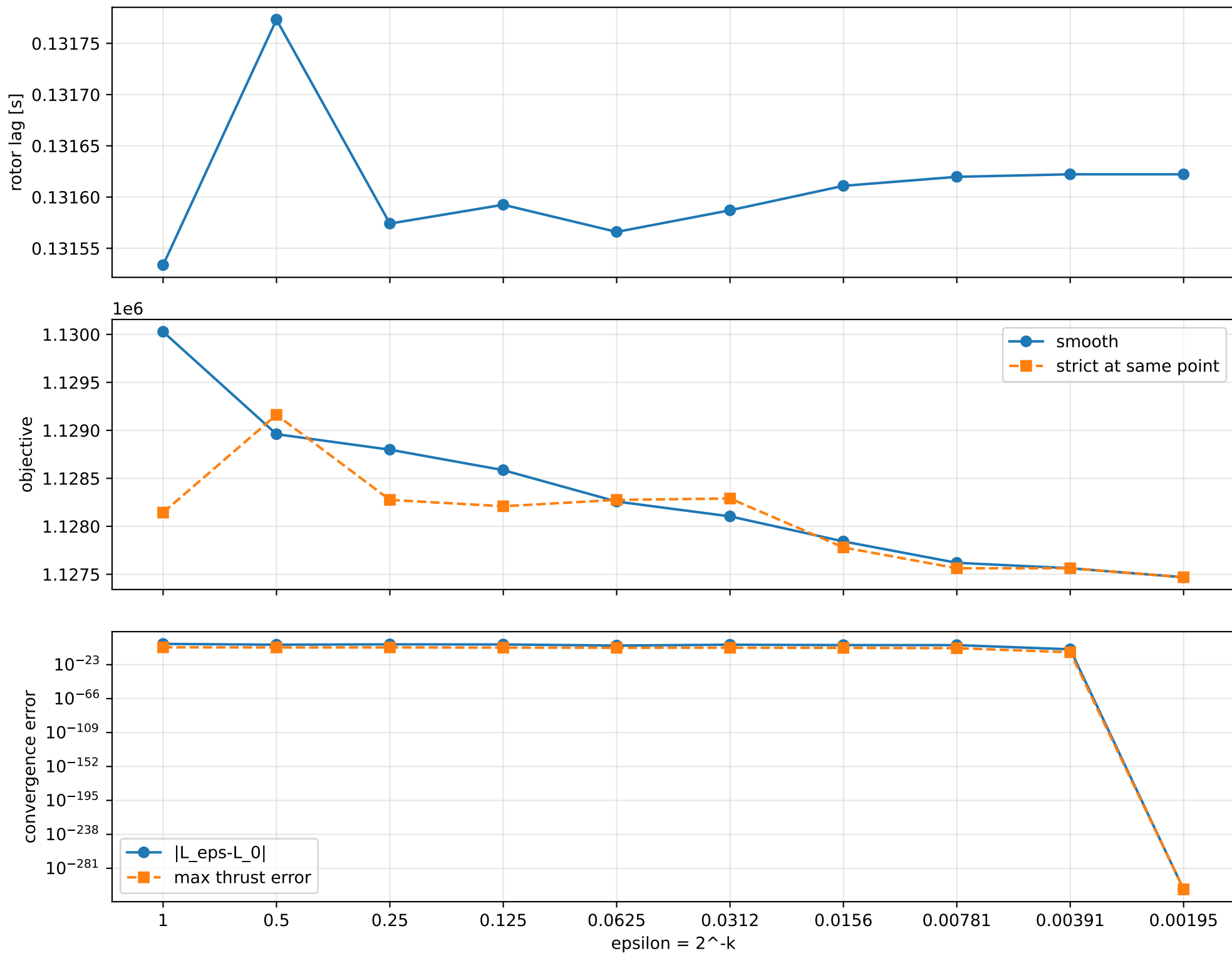
gyro [rad/s]



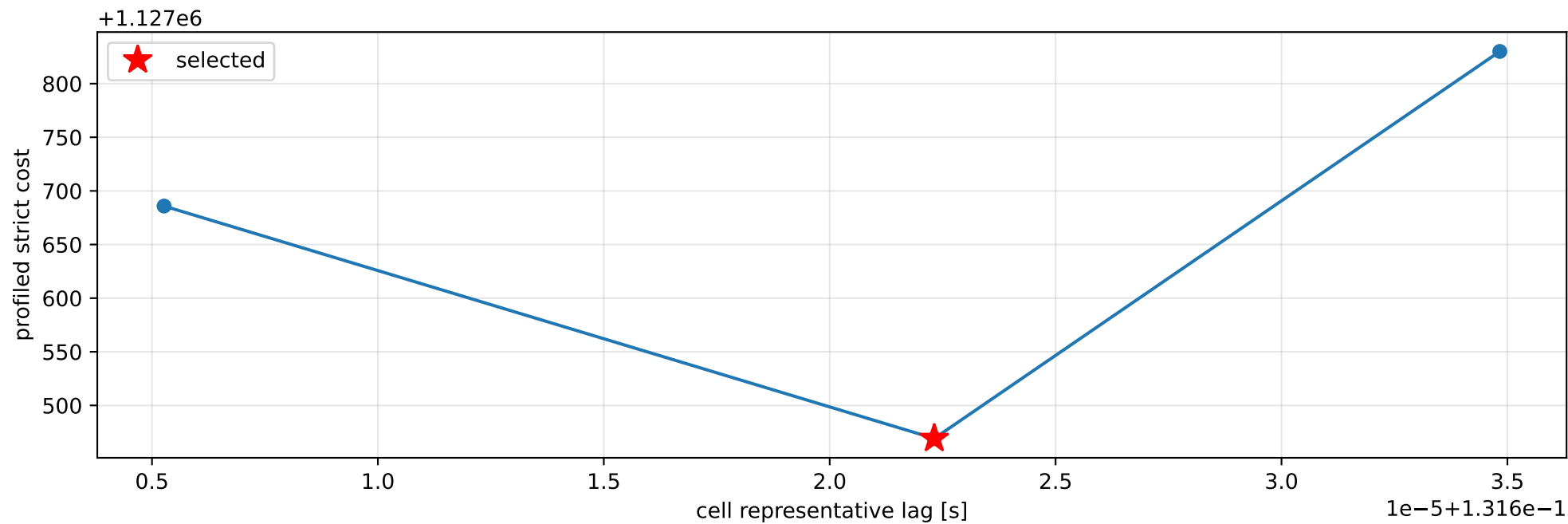
specific force [m/s²]



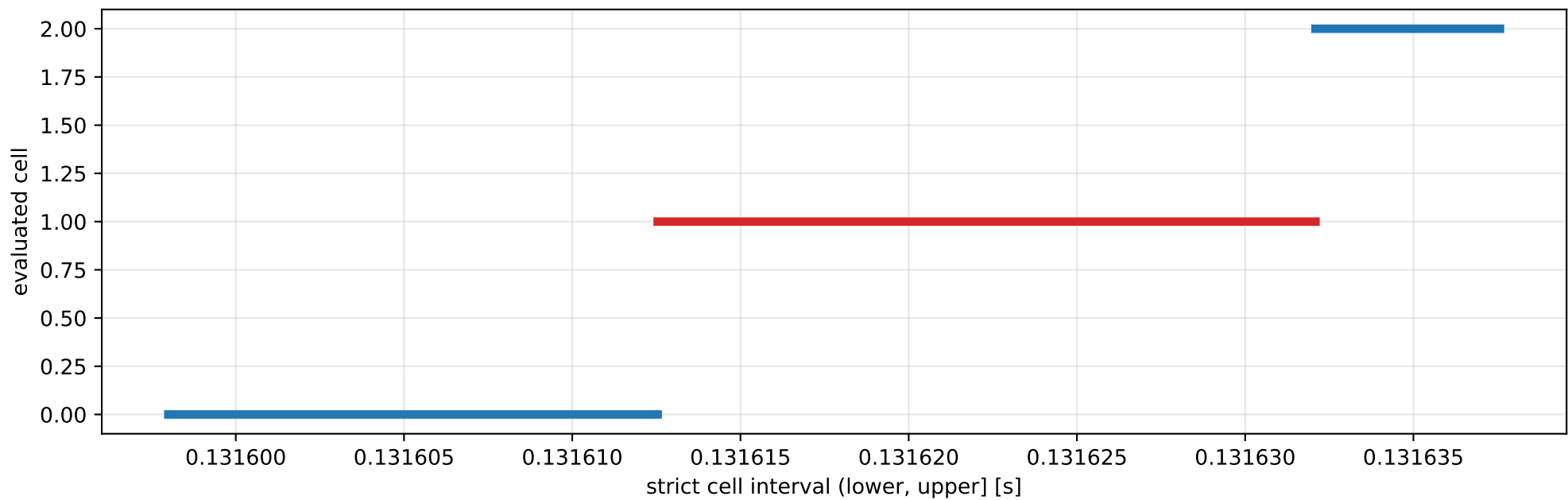
cov_diagonal: smooth-to-strict continuation



cov_diagonal: exact strict-ZOH lag cells



1e-5+1.316e-1



selected (0.131612539, 0.13163209] s; final smooth 0.1316222 s; data boundary=False

cov_diagonal: parameter estimate

Case: cov_diagonal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.29161947

inertia [kg m²]:

```
[[ 0.09813283 -0.00237684  0.02412945]
 [-0.00237684  0.13615278 -0.03288137]
 [ 0.02412945 -0.03288137  0.16159506]]
```

CoG body [m]: [0.00563654 0.05095243 -0.05177574]

force effectiveness: [0.36670911 0.43181078 0.56688361 0.53127274]

scale-free J/m [m²]:

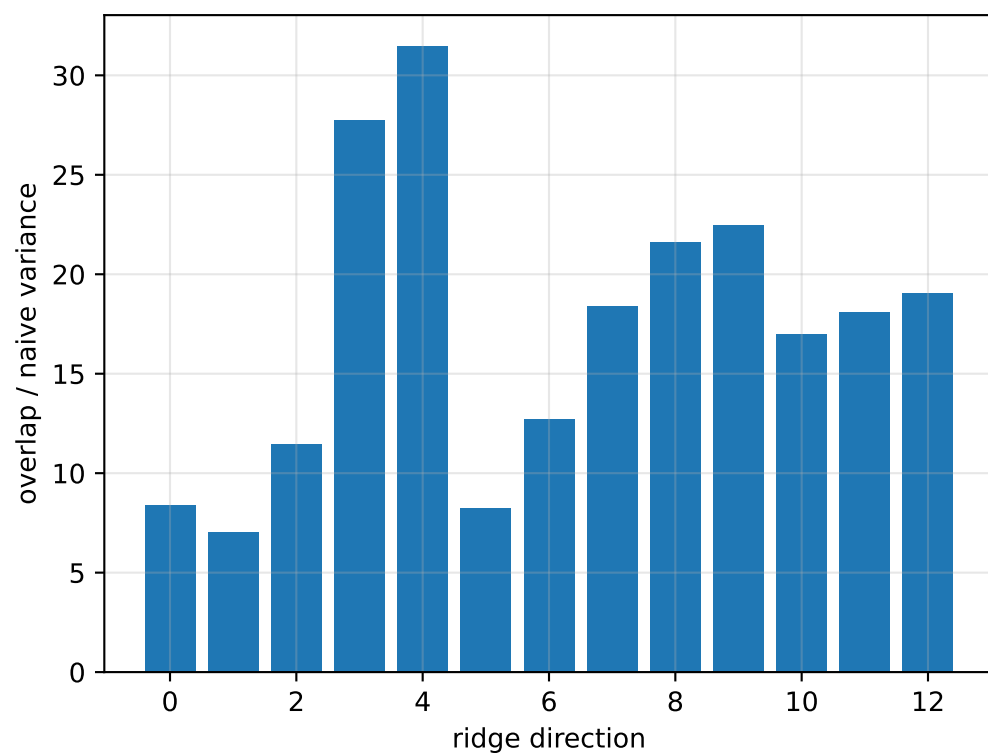
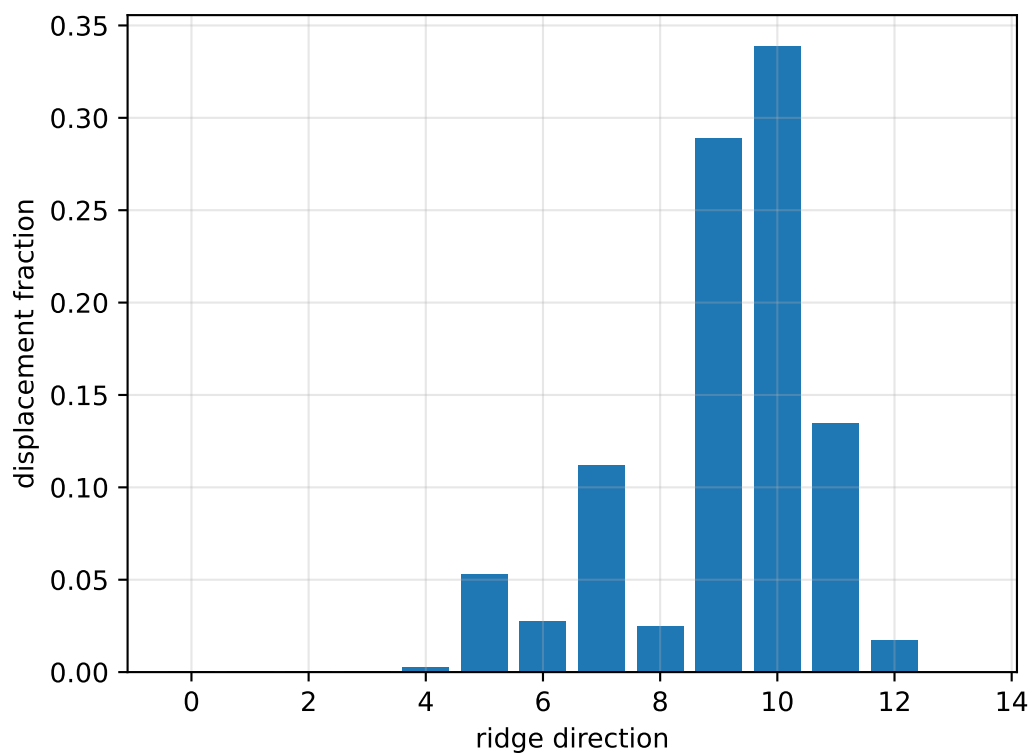
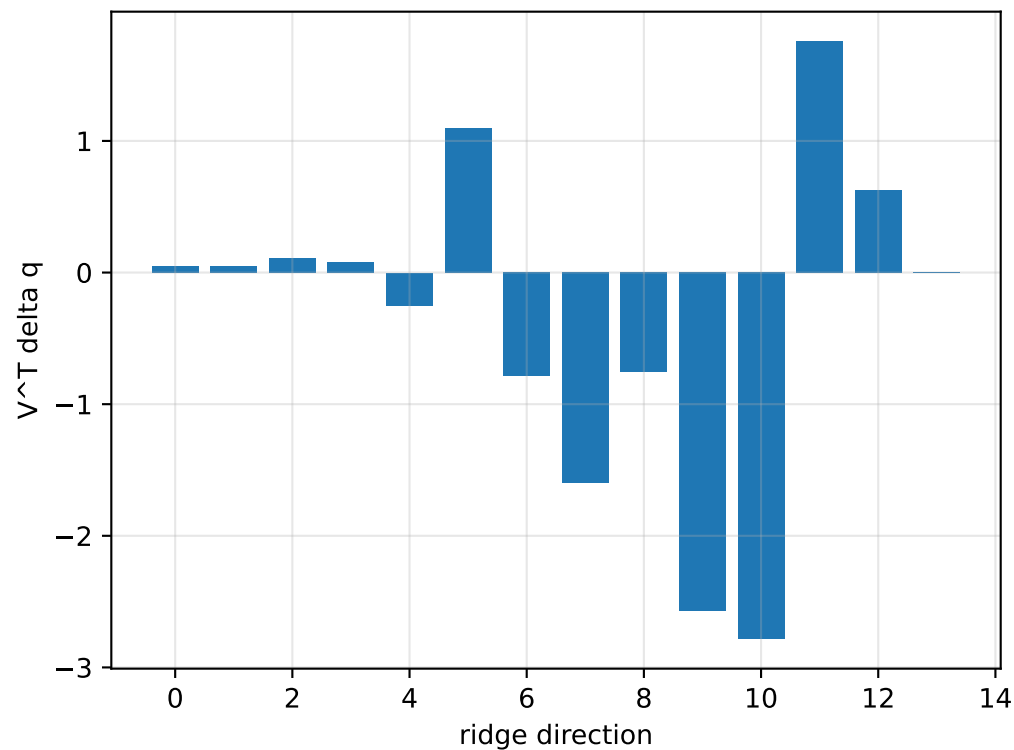
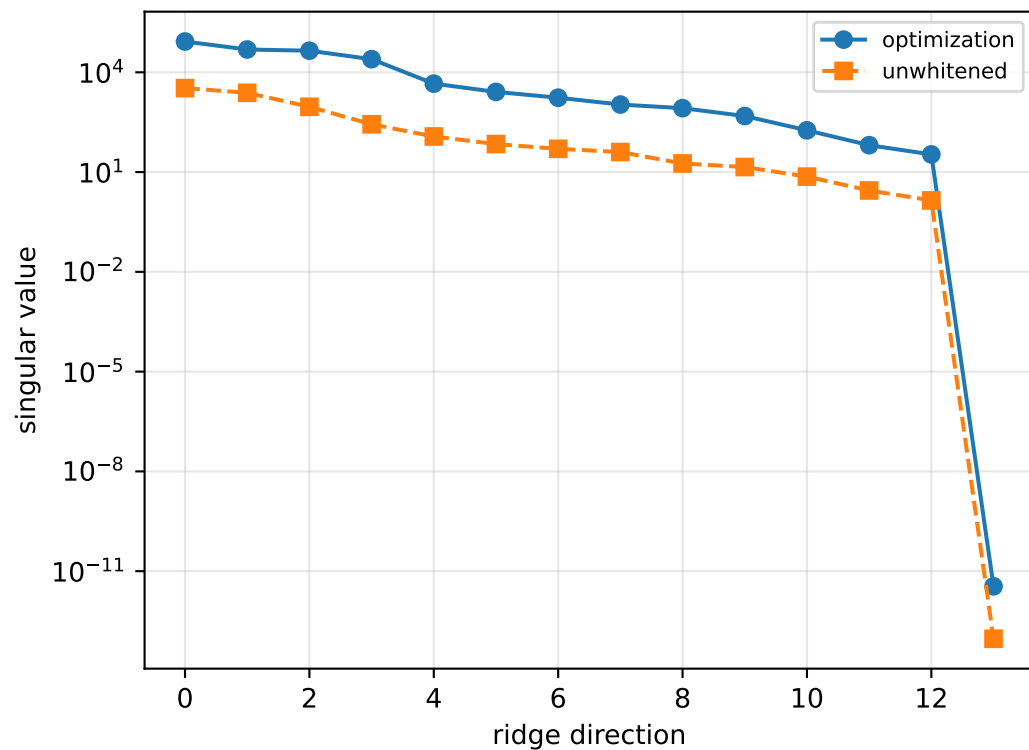
```
[[ 0.07597658 -0.0018402  0.01868155]
 [-0.0018402  0.10541246 -0.02545748]
 [ 0.01868155 -0.02545748  0.12511042]]
```

scale-free f/m: [0.2839142 0.33431734 0.43889367 0.41132296]

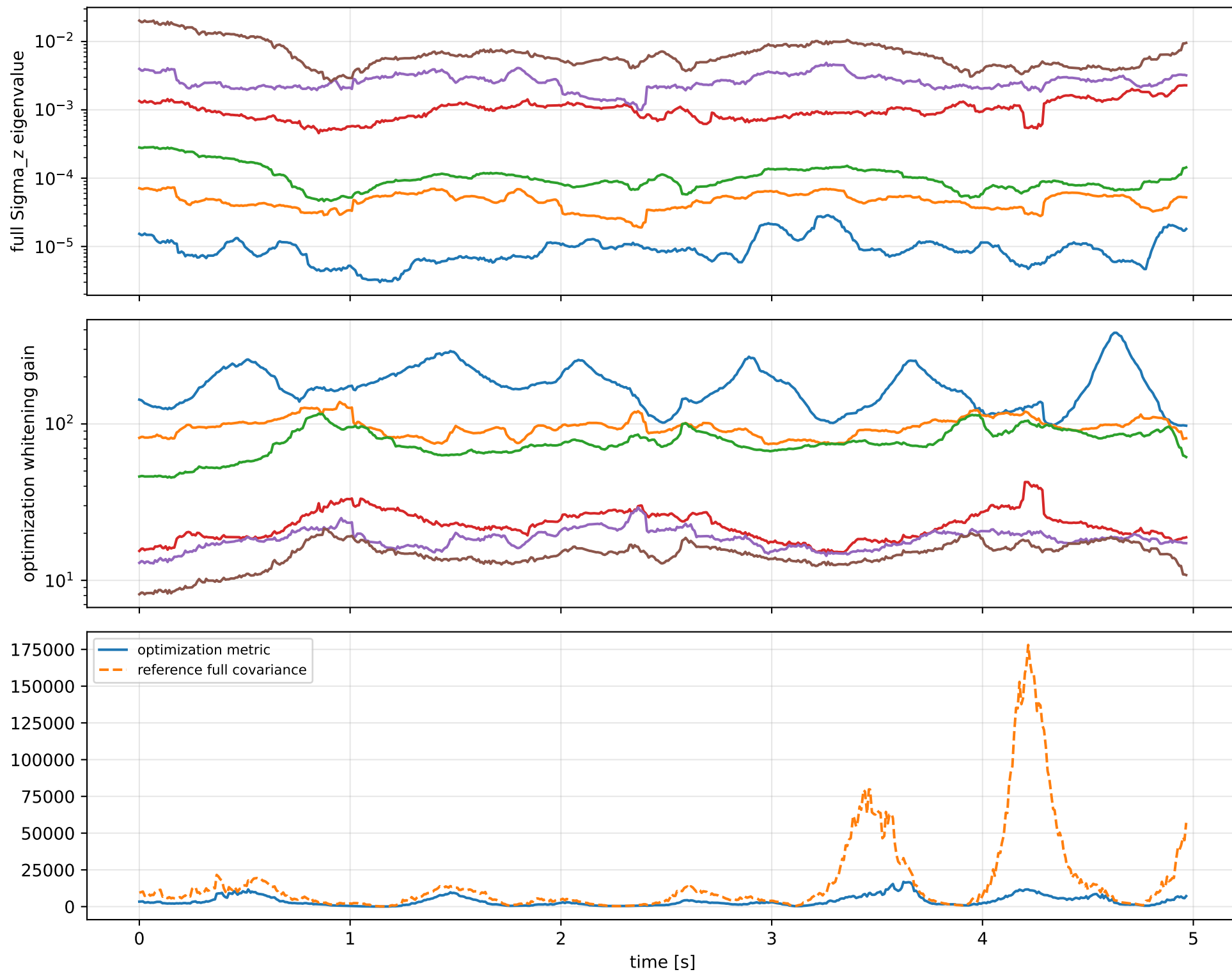
rotor lag cell: (0.131612539, 0.13163209] s

representative: 0.131622314 s

cov_diagonal: ridge spectrum (rank 13, nullity 1, $||\mathbf{v}||=6.7\text{e-}12$)



cov_diagonal: optimization vs reference covariance



cov_diagonal: wrench / closure (force max 3.89e-15, torque max 4.44e-16)

